5. (a) On the same axes, sketch the graphs of $y = x + 2$ and $y = x^2 - x - 6$ showing the coordinates of all points at which each graph crosses the coordinate axes.

(4)

(b) On your sketch, show, by shading, the region R defined by the inequalities

$$y < x + 2 \quad \textbf{and} \quad y > x^2 - x - 6$$

(1)

(c) Hence, or otherwise, find the set of values of x for which $x^2 - 2x - 8 < 0$

(3)